

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

SA3 - General Insurance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) You are working as a pricing actuary at medium sized general insurance company in India, Company A. This company writes direct motor, health, fire, liability business.

The Company is planning to expand in health insurance.

- i) Explain the characteristics of health insurance line of business. [8]
- ii) List the risk and rating factors that can be used to set premiums for individual health insurance and group health insurance. [5]
- iii) Describe with examples, the adjustments that may be made to the base experience to obtain burning cost premium. [15]

The Company is looking to develop a frequency model for its auto insurance.

- iv) Explain why it may be necessary for this company to adjust claims data for pricing in respect of large individual claims and catastrophes. [7]

As a starting point, the Company intends to construct a generalised linear model (GLM) for claim frequency.

For this initial version of the model, the Company proposes three parameters:

β_1, β_2 , and β_3

where:

β_1 : is parameter representing male policyholders,

β_2 is parameter representing vehicles located in Territory A, and

β_3 is the intercept, representing the baseline frequency level when the relevant indicator variables are not present.

Number of claims:

Gender	Territory A	Territory B
Male	700	600
Female	400	420

Number of exposures:

Gender	Territory A	Territory B
Male	1400	1000
Female	1000	1200

The objective is to use this GLM specification to estimate how frequency varies by gender (male vs non-male) and territory (Territory A vs Territory B), and to provide a sound basis for further model expansion and validation.

v) Define design matrix [X].

{Hint- design matrix is made up of $n+1$ columns in which 'n' columns are indicator (dummy) variables to capture the presence/absence of each category (or each non-base category), and one additional column is the base intercept term} [2]

vi) Define vector of responses [Y]. [2]

vii) Assuming $\beta_3 = 0.35$, solve a generalised linear model with normal error structure and identity link function for β_1 . Interpret the results of the model. [6]

viii) Someone in your team determines that the analysis would be improved by assuming a Poisson error structure with a log link function. Identify two reasons this structure may better suit this data. [2]

With the increasing use of artificial intelligence and machine learning in insurance pricing, GLM is increasingly being supplemented or replaced by more flexible methods. In this context, machine learning techniques which train on certain set of data provide a useful balance of strong predictive performance and practical explainability, since they capture nonlinear effects and interactions.

ix) How would you explain to underwriting team about how a GLM model provide better interpretability compared to machine learning techniques which train on a certain set of data? [5]

[52 Marks]

Q.2) A large general insurance company is rapidly expanding its operations. You are a reserving and capital modelling actuary in the company.

i) As a reserving actuary, explain briefly which technical reserves would need to be estimated and what would each of these reserves cater to. [7]

To understand the run-off of claims reserves better, you and your team are considering over-dispersed Poisson (ODP) method.

ii) List the key assumptions of ODP method. [2]

iii) Discuss the issues of ODP method around the following:

- asbestos claims
- sparse data in case of cyber liability insurance
- Reported claims may be subject to policy limits (right-censoring at the limit) or only claims above a deductible are recorded (left-truncation).
- A few catastrophe claims (hurricane, earthquake) are orders of magnitude larger than typical claims.

[8]

With the upcoming IND AS 117, an insurance company is planning to upgrade its current policy and claims system. In addition, much of the design and installation of the new system will be outsourced to a third party.

iv) Describe what is meant by “operational risk”. [2]

v) Identify potential sources of operational risk this project entails and suggest ways to mitigate each one. [12]

[31 Marks]

Q.3) You are an actuary working in the Analytics team of a fast-growing, digital general insurance company. The company writes a wide range of business. Given its rapid expansion and competitive pricing strategy, the company relies heavily on reinsurance to manage risk exposure and protect capital.

i) Outline some of the investigations you would carry out on reinsurance. [7]

The increasing claims ratio particularly in motor and health lines have raised concerns around claims volatility and potential fraud. Senior management has therefore asked the analytics team to strengthen fraud prevention frameworks.

ii) Suggest measures that could help control instances of fraudulent policyholder conduct. [10]
[17 Marks]
